

Jianghua Wu

Research Scientist, Shenzhen Research Institute of Big Data (SRIBD)

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◆ Education

- **Ph.D.**, Electrical and Computer Engineering Aug. 2023
University of Connecticut (UConn), Storrs, CT, USA
Dissertation topic: *Synergistic Integration of Machine Learning and Mathematical Optimization in Power System Planning and Operations*
Advisor: Peter B Luh, IEEE Life Fellow. <https://www.ee.uconn.edu/peter-b-luh/>
- **B.S.**, Building Electricity and Intelligence June 2018
Shenyang Jianzhu University (SJZU), Shenyang, China

◆ Research/Work/Internship Experience

- *Research Scientist*, Shenzhen Research Institute of Big Data (SRIBD), China Sept. 2023 - Current
 - Develop high-performance optimization solvers for power system operations
 - Research on Learn-to-Optimize algorithms for mixed integer programming (MILP) problems
 - Research on Large Language Model (LLM) assisted mathematical modeling and optimization, and relevant applications on power and energy networks
 - Research on the implementation and operation of virtual power plants (VPPs)
 - Research on the power system planning and operations under vehicle-to-grid (V2G)
- *Research Assistant*, University of Connecticut (UConn) Sept. 2018 - Aug. 2023
 - Research and develop optimization algorithms in power system planning and operations
 - Investigate and explore the synergistic integration of machine learning and mathematical optimization in mixed integer programming problems (MILP)
- *Intern*, Midcontinent Independent System Operator (MISO), Carmel, IN, US May 2020 - Aug. 2020
 - Develop an optimization algorithm on large-scale market clearing optimization problems and uncertainty analysis
 - Improve the feasibility of operation decisions and reduce the problem-solving time
- *Intern*, Midcontinent Independent System Operator (MISO), Carmel, IN, US July 2019 - Aug. 2019
 - Explore and assess optimization algorithms on large-scale Unit Commitment
 - Assist in developing High-Performance Power-Grid Optimization (HIPPO) platform, and integrate several algorithms into HIPPO
- *Research Assistant*, Shenyang Jianzhu University Sept. 2015 - June 2017
 - Validate the Kalman filter-based fault detection method for HVAC under cold weather, and assist in the development of the corresponding fault detection system

◆ Publication

Journals & CCF-A

- [1] J. Xiong, Y. Huang, Y. Wang, L. Yang, J. Wu, S. Lei, A. Wang, "Successive Fixing for Large-Scale SCUC Using First-Order Methods", *Electric Power Systems Research*, 2026. Available at SSRN 5101560. Available with doi 10.1016/j.epsr.2026.113414.
 - [2] **J. Wu***, Y. Chen, B. Yan, and M. A. Bragin, Z. Wang, "A Decomposition-Based Learn-to-Optimize Approach with Feasibility Layer Assistance for Sub-Hourly Unit Commitment," *Electric Power Systems Research*, 2026. Available with doi 10.1016/j.epsr.2026.113248.
 - [3] D. Li, J. Wu, C. Xiao, Q. Shi,, "Enhanced Non-Intrusive Load Monitoring through Tensor- Based
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Encoding and Involution Networks,” IEEE Internet of Things, Mar. 2026.

- [4] Y. Liu, Z. Tang, Y. Wang, **J.Wu***, “Hierarchical Optimization for Cross-regional Planning and Scheduling of Hydrogen Energy Storage Systems Considering Wind Uncertainty,” *Journal of Energy Storage*, Volume 125, 2025, 116669.
- [5] Q. Chen, L. Li, Q. Li, **J. Wu**, A. Wang, R. Sun, X. Luo, T. Chang, Q. Shi. “When GNNs meet symmetry in ILPs: an orbit-based feature augmentation approach,” *ICLR*, 2025.
- [6] W. Lin, Y. Wang, **J. Wu**, F. Feng, “Model-Free Aggregation for Virtual Power Plants Using Input Convex Neural Networks,” *IEEE Transactions on Smart Grid*, vol 16, pp 2404 – 2415, 2025.
- [7] **J. Wu***, P. B. Luh, Y. Chen, B. Yan, and M. A. Bragin, “Synergistic Integration of Machine Learning and Mathematical Optimization for Unit Commitment,” *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 391-401, 2024.
- [8] W. Lin, J. Hua, H. Jian, J. Xue, **J. Wu**, C. Wang, and Z. Lin, “High-dimension tie-line security regions for renewable accommodations,” *Energy*, vol. 270, May. 2023.
- [9] **J. Wu***, P. B. Luh, Y. Chen, M. Bragin, and B. Yan, “A Novel Optimization Approach for Sub-Hourly Unit Commitment with Large Numbers of Units and Virtual Transactions,” *IEEE Transactions on Power Systems*, vol. 37, no. 5, pp. 3716–3725, Sep. 2022.
- [10] L. Sun, **J. Wu**, H. Jia, and X. Liu. “Research on Fault Detection Method for Heat Pump Air Conditioning System under Cold Weather,” *Chinese Journal of Chemical Engineering*, vol. 25, pp. 1812-1819, 2017.

Conferences

- [1] Z. Zhu, Z. Zhou, Y. Wang, Y. Liu, **J. Wu***, “Temporal Decomposition with Selective Fixing and Relaxation for Fast Distributionally Robust Unit Commitment”, 2026 IEEE 2nd International Symposium on the Application of Artificial Intelligence in Electrical Engineering (AAIEE), Beijing, China, April 2026.
- [2] D. Dong, **J.Wu**, Y. Liu, “A Game-Theoretic and Learning-Based Framework for Flexible Virtual Power Plant Operation under Dynamic Electricity Markets”, 2026 IEEE Power & Energy Society International Meeting (PESIM), Hong Kong, China, Jan. 2026.
- [3] B. Tang, X. Ren, P. Xiao, S. Lei, X. Sun, **J. Wu***, “A Hybrid Autoencoder-Transformer Model for Robust Day-Ahead Electricity Price Forecasting under Extreme Conditions,” 2025 IEEE International Symposium on the Application of Artificial Intelligence in Electrical Engineering (AAIEE), Beijing, China, April 2025.
- [4] Z. Tang, Y. Wang, Y. Liu, **J.Wu***, “Hierarchical Optimization for Cross-regional Planning and Scheduling of Hydrogen Energy Storage Systems Considering Uncertainty,” *2024 International Conference of Electrical, Electronic and Networked Energy Systems (EENES)*, Xi’an, China, 2024.
- [5] Y. Wang, P. Ou, **J.Wu***, “A Novel Price-Oriented Supply-Demand Interaction Scheduling Model Optimizing Virtual Power Plant Operations”, 2024 IEEE PES 16th Asia-Pacific Power and Energy Engineering Conference (APPEEC 2024), Nanjing, China, 2024.
- [6] P. Ou, Y. Wang, W. Lin, **J. Wu***, “An LLM-Based Modeling and Decision Optimization for User-Centric Electric Vehicle Charging,” 2024 IEEE Conference on Energy Internet and Energy System Integration (IEEE EI² 2024), Shenyang, China, 2024.
- [7] M. Xu, S. Lei, C. Ma, H. Liu, **J. Wu**, C. Peng, “Incentives for Energy Communities in Backup Energy Storage Investments for System Resilience Enhancement,” 2024 IEEE Conference on Energy Internet and Energy System Integration (IEEE EI² 2024), Shenyang, China, 2024.
- [8] **J. Wu***, P. B. Luh, Y. Chen, B. Yan, and M. A. Bragin, “A Decomposition and Coordination Approach for Large Sub-hourly Unit Commitment,” *2020 IEEE Power & Energy Society General Meeting*

(PESGM), Montreal, QC, Aug. 2020.

- [9] L. Sun, **J. Wu**, L. Zhang, and Y. Yu. “Research on Fault Detection Method for HVAC under Cold Weather Conditions: A Case Study on Sino-German Building in Northeastern China.” *Chinese Process Control Conference*, Lanzhou, China, 2016.

Under Review

- [1]
- [2] B. Tang, Y. Zeng, X. Ren, P. Xiao, Y. Zhao, R. Lee, **J. Wu***, “COTN: A Chaotic Oscillatory Transformer Network for Complex Volatile Systems under Extreme Conditions,” *IEEE Transactions on Neural Networks* (under review)
- [3] Z. Zhou, Y. Wang, **J. Wu***, “A Fragmentation-Inhibiting Framework for Efficient Structure Detection in large-scale MILPs”, *INFORMS Journal on Optimization* (under review)
- [4] J. Xiong, L. Yang, Y. Wang, Y. Huang, **J. Wu**, S. Lei, A. Wang, “Relax-and-Cut for Temporal SCUC Decomposition”, *IEEE Transactions on Power Systems* (under review)
- [5] M. Xu, S. Lei, C. Ma, H. Liu, **J. Wu**, C. Peng, “Multi-Stage Coordination Between a Distribution System Operator-Multiple Energy Community Developers for Resilience Planning Using a Hybrid Game-Theoretic Approach”, *IEEE Transactions on Smart Grid* (under review)
- [6] W Liu, A Wang, D Ma, H Jiang, J Wu, W Yang, “Smoothing Binary Optimization: A Primal-Dual Perspective”, *ICLR 2026*. (under review)
- [7] Z. Liu, L. Zhang, Q. Li, **J. Wu**, G. Zhu, “Invar-RAG: Invariant LLM-aligned Retrieval for Better Generation,” arXiv preprint arXiv:2411.07021, *ICLR, 2026*. (under review)
- [8] Z. Liu, Z. Zhu, **J. Wu***, “Toward a Unified Review of Stepsize Policies across Machine Learning and Combinatorial Optimization”, *INFORMS Journal on Optimization* (under preparation)
- [9] L. Yang, B. Li, T. Ding, **J. Wu**, A. Wang, Y. Wang, J. Tang, R. Sun, X. Luo. “An Efficient Unsupervised Framework for Convex Quadratic Programs via Deep Unrolling,” arXiv preprint arXiv:2412.01051.

◆ **Contents**

- 深圳市大数据研究院, 吴江华, 史清江, 周润东, “电力机组组合的运行参数预测方法、装置、设备、介质”, 发明专利, 中国 (已授权)
- 深圳市大数据研究院, 张亮, 彭芮琪, 吴江华, 王阿康, “基于对抗时空网络的新能源功率鲁棒预测 方法、装置 ”, 发明专利, 中国 (已授权)
- 深圳市大数据研究院, 吴江华, 史清江, 周润东, “机组预测状态校验方法、装置、电子设备、存储介质”, 发明专利, 中国 (已受理)
- 深圳市大数据研究院, 杨林鑫, 李秉恒, 丁添, 吴江华, 王阿康, 孙若愚, 罗效东, “基于原始对偶混合梯度算法展开的信号处理方法、装置、设备与介质”, 发明专利, 中国 (已受理)

◆ **Projects**

- Digital employees of LLMs for elderly care and healthcare, Jinan Lixia Holding Group Co., Ltd. (On progress.)
- Non-intrusive load decomposition technology for high-power equipment on drilling platforms, China National Petroleum Corporation. (On progress.)
- Automatic Compilation of Subway Operation Schedules and Operational Optimization. CRRC Zhuzhou Institute Co., Ltd. (On progress.)
- Multi-Train Coordinated Regenerative Braking and Speed Level Adjustment Energy Saving. Qingdao Metro. (On progress.)

- “An accelerated decomposition and coordination technique through time-series prediction for unit commitment”, supported by Shenzhen Research Institute of Big Data (SRIBD), 02/2024 - 02/2025. (¥100,000)
 - Combining time-series prediction with the decomposition and coordination technique to provide fast & feasible solutions for unit commitment
 - “Machine learning-based optimization for sub-hourly unit commitment”, supported by Midcontinent Independent System Operator, 12/2020 - 12/2022. (\$400,000)
 - Investigated the research status of Mathematical Optimization and Machine Learning in Unit Commitment
 - Integrated Machine Learning into traditional Mathematical Optimization for hourly and sub-hourly Unit Commitment
 - Developed hybrid ML and optimization algorithms capable to deliver fast & feasible results
 - “Unit Commitment with Parallel SAVLR and Formulation Tightening”, supported by Midcontinent Independent System Operator, Grant No. 080914, 09/2020 - 5/2022. (\$400,000)
 - Synergistically combined SAVLR (an optimization algorithm) and formulation tightening to obtain efficient solving capability for the Unit Commitment problem
 - Accommodated the Ordinal Optimization concepts into SAVLR for a drastic reduction in computational time
 - Improved the performance of High-Performance Power-Grid Optimization (HIPPO) platform by integrating several efficient optimization algorithms and heuristics.
 - National Science Foundation Project “Empowering Smart and Connecticut Communities through Programmable Community Microgrids”, Grant No. ECCS-1831811. 2018492, 09/2018 – 04/2020. (\$800,000)
 - National Science Foundation Project “A Revolutionary Way to Tackle Challenging MILP Problems in Power Systems through Accelerated Convergence, Formulation Tightening and Asynchronous Optimization”, Grant No. ECCS-1810108, 08/2018 – 05/2023. (\$360,000)
 - “Stochastic Sizing and Operation of Grid-Level Energy Storage Systems (ESSs) under Intermittent Renewable Generation and Increasing Load Forecasting Uncertainties”, supported by Brookhaven National Laboratory, 08/2018 – 08/2019.
 - Developed probabilistic models for wind generation variability
 - Formulated stochastic operation problem with given ESSs capacities
 - National Natural Science Foundation of China Project “Research on Main and Accessorial Equipment Scheduling Method for the Steelmaking-Continuous Casting Process Based on Effective Lagrange Relaxation Iterative Algorithm”, Grant No: 61503259, 2017 - 2018.
 - Assisted in the development of the Lagrange Relaxation-based model for the equipment scheduling problem
 - Validated the cost-effectiveness of the developed model on the Steelmaking-Continuous Casting Process
 - Conducted a sensitivity analysis to identify the key factors that affect the performance of the algorithm such as equipment downtime or unexpected maintenance needs
 - China Postdoctoral Science Foundation Funded Project “Research on Batch Planning and Scheduling Integrated Optimization Method for the Steelmaking-Continuous Casting Process Based on Lagrange Framework”, Grant No: 2017M611261. 2016 – 2017.
 - Implemented the Lagrange-based optimization method in a software program, i.e., CPLEX
 - Compared the performance of the Lagrange-based optimization method with other existing
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methods in terms of quality, speed, and robustness

- “Hanyu” project of Shenyang Jianzhu University “Research and application of fault detection method for air conditioning”, Grant No: S121201318. 2015 - 2016.
 - Designed a Kalman filter-based fault detection algorithm for air conditioning systems
 - Validated the Kalman filter-based fault detection method for HVAC under cold weather
 - Assisted in the development of the corresponding fault detection system

◆ Reports

- 面向大规模安全约束机组组合问题的多种求解策略, 中国运筹学会数学规划分会常务理事会议暨数学优化研讨会, 福州, 中国, 11/2025
- Synergistic Integration of Machine Learning and Mathematical Optimization for Difficult Optimization Problems, Keynote Speech in 2024 International Conference of Electrical, Electronic and Networked Energy Systems (EENES 2024)
- Learning-to-Optimize Techniques and Applications on Complicated Planning and Scheduling problems. Seminar on Optimization Models and Algorithms for Power Systems (2024), Academy of Mathematics & Systems Science, Chinese Academy of Science, Beijing, China, 09/2024.
- Learning-to-Optimize Techniques and Applications on Power System Operations, EENES 2024, Xi'an, China, 10/2024.
- Synergistic Integration of Machine Learning and Mathematical Optimization for Unit Commitment, 2023 FERC Technical Conference Increasing Market and Planning Efficiency through Improved Software, FERC, Washington DC, USA, 06/27/2023 – 06/29/2023.
- Synergistic Integration of ML and Mathematical Optimization for Unit Commitment, 2022 FERC Technical Conference Increasing Market and Planning Efficiency through Improved Software, FERC, Washington DC, USA, 06/21/2022 – 06/23/2022.
- A Novel Optimization Approach for Sub-Hourly Unit Commitment with Large Numbers of Units and Virtual Transactions, 2020 IEEE Power & Energy Society General Meeting (PESGM), Montreal, QC, Canada, 08/02/2020 – 08/06/2020
- A Novel Optimization Approach for MILP Problems, 2020 FERC Technical Conference Increasing Market and Planning Efficiency through Improved Software, FERC, Washington DC, USA, 06/23/2020 – 06/25/2020.

◆ Skills & Expertise

- Math Modeling; Operational Research Optimization; Mixed Integer Linear Programming; Nonlinear Programming; Dynamic Programming; Game Theory; Machine learning; Deep learning
- Software & Programming: PYTHON; GUROBI; ILOG CPLEX; C; MATLAB

◆ Honor & Awards

- Outstanding Keynote Speech in 2024 International Conference of Electrical, Electronic and Networked Energy Systems (EENES 2024)
 - Third Prize in the Competition of the Power Network Optimization Solver, the Ministry of Industry and Information Technology, China 2023
 - Doctoral Dissertation Fellowship 2023
 - UConn Conference Participation Fellowship 2022
 - Honorable Mention in the Interdisciplinary Contest in Modeling 2017, 2016
 - National Scholarship for Encouragement 2017
 - Excellent graduates in Liaoning Province 2017
 - Outstanding student scholarship of Shenyang Jianzhu University 2017, 2016, 2015
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- Model Student of Academic Records of Shenyang Jianzhu University 2017, 2016, 2015
- “Merit Student” of Shenyang Jianzhu University 2017, 2016, 2015

◆ **Affiliations**

- CAAI (Life member)
 - ORSC (Life member)
 - Member of IEEE, IEEE Power & Energy Society, and IEEE Young Professionals, (2018 - present)
 - Member of IEEE Consultants Network (2023 - present)
 - Co-Chair of the special session “Artificial Intelligence and Big Data Technologies for Low-carbon Power Systems” in 2024 International Conference of Electrical Electronic and Networked Energy Systems, EENES 2024
 - Journal Reviewer of
 - IEEE Transactions on Power Systems (TPWRS)
 - Applied Energy
 - IEEE Access
 - IEEE PES letters
 - Sustainable Energy, Grids and Networks
 - Journal of Advanced Manufacturing Technology (JAMT)
 - International Journal of Simulation and Process Modelling (IJSPM)
 - Conference Reviewer of
 - IEEE Power & Energy Society General Meeting (PESGM)
 - IEEE Conference on Energy Internet and Energy System Integration (EI2)
 - ICLR
 - AAAI
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